



If Saarinen Were Designing Dulles Today

An Analytical Report Generated by the AI Research Council

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Abstract

Eero Saarinen's IAD encoded a set of design principles — about modal separation, monumental civic architecture, flexible mid-field expansion, and a deliberate passenger experience — that were the deeper intent of the 1958–62 design, separable from its visual signature. Some of those principles have aged into operational liabilities: the original mobile lounges, the fixed terminal geometry, the sterile-to-non-sterile transitions. Others remain quietly correct and have been violated by every subsequent expansion: modal clarity, the single-arrival-experience, mid-field flexibility. A 2026 reinterpretation faithful to Saarinen's design philosophy — not his aesthetic — would make materially different choices than the current MWAA master plan in specific, identifiable ways. The report examines this claim across the following: A sophisticated reader cannot dismiss the piece as nostalgia in the first 500 words — the catalog of principles has to be specific, sourced, and separable from aesthetic; At least one Saarinen principle the piece concludes is obsolete must be argued at full strength — no hagiography; The contrarian's case (that the entire framing romanticizes a single architect and a single moment, and that current best-practice airport design has already absorbed what is true and discarded what is not) is presented strongly enough that the reader feels its force; The 2026 reinterpretation lands as five or six specific design-philosophy moves that a MWAA staffer reading it can hold against the current master plan and identify which decisions are at stake; The constraint-compatibility question (Saarinen-philosophy design vs. 2026 TSA/FAA/ADA/airline-commercial reality) is answered concretely — naming which constraints are worth fighting and which require letting go, not punting.

How This Report Was Produced

This report is the product of the Transform Airports AI Research Council — a multi-agent analytical system running on Anthropic's Claude models. Rather than a single model writing in one pass, the Council convenes a panel of specialized research agents, subjects their work to adversarial review, and verifies every claim before publication. The process runs in four stages, with human review at two checkpoints.

Stage 1 — Independent research. Each research agent below produced an evidence brief without seeing the others' work, so the analysis draws on genuinely independent lines of evidence rather than a single line of reasoning.

Stage 2 — Synthesis and adversarial review. A Strategist synthesized the briefs into an argument; a Red Team attacked it across two rounds; the Strategist revised after each.

Stage 3 — Edit and fact-check. An Editor tightened the prose. A Fact-checker verified every number against the research briefs, removing or flagging what it could not source.

Stage 4 — Production. The verified draft was formatted into this document.

Research agents seated for this report

- Airline Commercial Strategist
- Airport CEO
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- Architectural Historian
- Aviation Historian
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- Infrastructure Economist
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Process agents (on every report)

- **Strategist** — synthesized the research briefs into a single argumentative draft.
- **Red Team** — attacked each draft to surface weak claims and logical gaps.
- **Editor** — tightened the prose and removed filler without adding new content.
- **Fact-checker** — verified every numerical and attributed claim against the briefs.

Each agent is governed by a versioned specification — its analytical mandate, evidentiary standards, source discipline, and argumentative posture. The complete technical definitions for every agent on the Council are published and openly available at:

<https://github.com/TransformAirports/ai-council/tree/main/.claude/agents>

The Stopwatch and the Hammock: Saarinen's Dulles, Read Honestly for 2026

A position paper on what a faithful reinterpretation of Saarinen's design philosophy would, and would not, ask of MWAA's \$22 billion revitalization program.

I. Chantilly, 1959

In the year before he drew the curve that would become the most photographed roof in American aviation, Eero Saarinen carried a stopwatch. So did his staff. They flew to other airports and timed themselves walking from curb to gate, and they timed every passenger they could shadow without being asked to leave. They came back with a number: 900 feet was the average walk at comparable airports of the period. The Civil Aeronautics Administration, which had selected Saarinen in May 1958 alongside Ammann & Whitney, had calculated that a conventional terminal sized to the projected build-out at Chantilly would require concourses extending more than 8,000 feet — a mile and a half of finger pier, end to end. The Saarinen office's response was an engineered substitution: bring the aircraft to the passenger by vehicle, and keep the building short enough that 900 feet became 200. Saarinen's own statement on the project, published the year he died: "Sometimes the problem and the times are right for an entirely new functional approach to a problem. No one asked us to grapple with the problem of the jet age terminal beyond the question of pure architecture, but I believe the architect has to assume that kind of responsibility."

The catenary roof — 1,792 precast concrete panels, each weighing 800 pounds, suspended on inch-thick cables strung between 65-foot inward-sloping piers — was not the argument. It was the conclusion. The argument was the stopwatch work that preceded the drawing. The roof's structural logic, monolithic and pre-stressed by the panel dead load, would later become the inheritance constraint that defines what is and is not possible inside the building it makes. The conclusion is also the cage.

That two-sided fact is what the present conversation about the \$22 billion revitalization of Dulles has lost. The thesis of this paper is narrow. **The \$22 billion plan now in front of MWAA honors Saarinen's operational logic at the level of modal coherence and is poised to repeat the documented pattern of every IAD expansion since 1996 at the level of execution — preserve the form, abandon the intent.** The plan is more faithful to the rescuable Saarinen than the aesthetic preservationists recognize. It is also at risk of becoming what every PIT-style anchor-carrier hub terminal becomes when the anchor carrier's strategy shifts. The Saarinen-philosophy frame names both risks at once.

What follows gives MWAA's program staff six specific moves they can hold against the master plan, an honest account of what the structural-engineering inheritance forbids, and a verdict on the Pittsburgh comparison that the plan owes its board and has not yet delivered.

II. Executive Summary

1. **Saarinen's IAD design encoded three principles documented in the primary record, and one operational consequence the strategist frames as a fourth.** The architectural-historian's documentary read enumerates three: a compact single terminal; midfield apron flexibility; and grade-separated passenger movement. Modal clarity — the passenger sequence reading as a single coherent thing — is the experiential consequence of the other three, not an independently sourced principle. This paper treats it as an analyst construction synthesized from the architectural and operational record, not as a fourth Saarinen principle.
2. **The mobile lounge concept "operated as designed for 23 years. It has been operationally superseded for 40. The duration of the failure now exceeds the duration of the success."** Saarinen's own writing makes the dependency explicit: "The acceptance of the mobile lounge concept allowed us to make the terminal a single compact building." When the vehicle failed at hub-bank volumes after 1985, the compact-terminal principle had to fail with it.
3. **The compact-single-building principle is not merely outgrown but structurally unrecoverable inside the inherited envelope.** The catenary structure is monolithic; the 150-foot column-free interior over a 1,240-foot footprint cannot be subdivided, the column grid cannot be modified, and the perimeter cannot be reconfigured without breaking pre-stress equilibrium. Restoring a compact single-terminal experience at 45 million annual passengers is, in chief-engineer terms, a new-building question stated in the language of renovation.
4. **The November 10, 2025 Concourse D mobile lounge collision injured eighteen passengers, one of whom lost consciousness.** A second collision followed later the same month. The two-collision sequence ends the heritage argument independent of any design discussion. The fleet's original manufacturers no longer supply parts; MWAA approved a \$16.4 million contract in 2023 for two prototype refurbished Plane-Mates with a \$143 million option on the balance of the fleet, much of which has not yet been exercised. The 2026 question is not whether to retire mobile lounges. It is what the principle they served looks like, executed in 2026 idiom.
5. **MWAA's debt service coverage sat at 3.16x in 2024 and Moody's projects it compressing toward 1.3x against a 1.25x covenant floor through the peak debt-service years.** Independent analysis projects cost per enplanement reaching \$90.64 by 2035 under the \$22 billion plan, against a current \$12.88 and a level — \$26.47 — at which United walked away from IAD as a hub the first time. No design philosophy survives a CPE that makes the airport uncompetitive for the carrier whose roughly 70 percent landed-weight share gives it functional veto authority through the Use and Lease Agreement's Majority-in-Interest mechanism.
6. **The \$22 billion plan designs four linear concourses specifically around United's connecting-bank requirements and so makes IAD more United-dependent, not less.** Pittsburgh International built a \$1 billion hub terminal in 1992 designed around US Airways' 80% market share; US Airways dehubbed in 2004

and enplanements fell 57% by 2013, with Concourse E becoming surface parking. The verdict this paper commits to: IAD's exposure is structurally lower than PIT's was — by virtue of the Lufthansa joint venture, the DCA slot constraint, and the diplomatic-gateway role — but lower is not low, and the four-linear-concourse design moves IAD in the wrong direction on this risk. The plan should be hedged. Specific hedging moves appear in Section VI.

7. **The TSA sterile-area wall is the binding architectural constraint**
biometrics will not dissolve on the time horizon that matters. TSA's Checkpoint Requirements and Planning Guide (May 2025) still requires walls adjacent to non-sterile areas to be a minimum of 8 feet high. At every named deployment reviewed, biometric systems sit on top of existing checkpoint architecture, not in place of it. The identity-verification step accelerates; the wall does not move.
8. **The project-specific Section 106 Programmatic Agreement is the live legal lever and applies to multiple program elements.** The Virginia State Historic Preservation Officer has determined that a project-specific PA is appropriate for the current terminal/concourse redevelopment EA. The PA can embed design-quality obligations — spatial, experiential, procedural — in a legally binding document if MWAA negotiates for them. It is the relevant instrument for the 300-foot terminal extensions, for the AeroTrain station dwell spaces, for the design relationship between the preserved main terminal and the new concourses, and for the spatial expression of the sterile-area boundary. The window closes when MWAA's board commits a bond issuance to a fixed scope.

III. The Argument

The stopwatch was the method, not the motif

Saarinen did not approach the Dulles commission from architecture. He approached it from measurement. The 900-foot average walk was a data point his office produced, not a polemical figure they borrowed. The 8,000-foot conventional alternative was the CAA's number, examined against more than 100 proposals before the mobile lounge scheme was selected in January 1960. The catenary roof followed the operational diagram — not the other way around.

What the documentary record attributes to the Saarinen office is specific and bounded: more than a year of stopwatch-timed passenger-flow studies at comparable airports; a 900-foot baseline established by direct observation; an internal review against more than 100 proposed terminal schemes coordinated with the FAA before selection in January 1960; and a sustained team composition pairing architectural staff with operational engineers from Ammann & Whitney. That is the empirical method as the briefs document it — not a vague invocation of "rigor" but a year of direct observation paired with quantitative comparison of program alternatives, paired with the willingness to let the building's form follow what the observation produced.

This matters because the public Saarinen — the Saarinen of the TWA terminal at JFK, who told a reporter five months before his death that if construction stopped at TWA the result would make "a beautiful ruin, like the Baths of Caracalla" — is an architect who optimized for emotional experience and trusted that operations would catch up. That Saarinen built TWA. That building closed entirely in 2001, thirty-nine years after it opened, having been operationally disqualified almost from the start: the design was completed before the 707 entered service, the terminal was too small by the time it was finished, and it limped through four decades of incompatibility with the aircraft it was supposed to serve.

The Saarinen who built Dulles was a different architect on the same payroll. He worked from the same office, the same year. He died in August 1961, age fifty-one, before either terminal opened. He produced two buildings simultaneously embodying opposite theories of what an airport is for. One has operated continuously for sixty-four years. The other became a hotel. The hagiographic literature has not done the work of separating them. The reader of this paper must.

What the stopwatch work produced, in the form that has the best claim on 2026, was three operational principles built into physical infrastructure. First, the terminal would be compact — small enough that a passenger would not lose orientation inside it. Saarinen wrote: "The acceptance of the mobile lounge concept allowed us to make the terminal a single compact building." Note the dependency. The compact terminal was not the principle. It was the consequence of choosing to bring the vehicle to the passenger. Second, the apron would be flexible — aircraft could park anywhere within lounge reach, any size at any position, without gate geometry constraining operations. Third, the passenger would never walk on the ramp. No fumes, no weather, no aircraft taxi paths to cross.

Each principle was independently defensible. Each was an engineering response to a documented problem. None required the specific vehicle Saarinen and the FAA settled on.

How the vehicle failed — and the principle it depended on failed with it

The numbers are pitiless. "The concept operated as designed for 23 years. It has been operationally superseded for 40. The duration of the failure now exceeds the duration of the success." Saarinen's original 1962 fleet of nineteen Chrysler/Budd vehicles, each carrying 102 passengers, could sustain throughput for a point-to-point international gateway handling 2 to 3 million passengers per year — Dulles's actual traffic through 1983. The failure threshold was not gradual; it was a cliff. When New York Air launched hub operations at Dulles in 1985 and traffic crossed 5 million, the connecting-bank structure required simultaneous multi-gate arrivals and departures that the lounge fleet could not serve without prohibitive queuing. By 1987, at 10 million passengers, midfield Concourses C and D had opened with conventional jetbridges and were absorbing the connecting bank traffic. The Saarinen operating diagram had been abandoned within 25 years of opening.

The cliff was structural, not mechanical. Each lounge served one departure or one arrival at a time, then returned to base for the next load. In a point-to-point operation with scheduled departure windows spaced across the day, this worked. In a hub bank with twenty or thirty simultaneous arrivals or departures, the fleet math collapsed. The wide-body introduction in 1971 — and the larger 150-passenger Plane Mate built specifically to handle the 747 — bought another decade. Then it ran out. The "temporary" Concourse C/D, opened in 1983 as a stopgap, is still hosting United's hub in 2026, forty-three years into its temporary life, with HVAC replaced, the sanitary system reconfigured after repeated flooding, and ceilings low enough to make a Monday-morning push feel like a submarine.

The contrarian surfaces a number that sharpens this further: in 1975, Dulles served 2.5 million passengers against a 6 million design capacity — 42% utilization — while DCA served 11.7 million. The mobile lounge airport ran below half capacity in the era it was supposed to dominate, in a market where its principal alternative ran at more than four times its traffic. The passenger-experience advantage that was supposed to distinguish the system did not, when passengers had every chance to choose it, produce a market preference. That is a fact the philosophical defense must answer for. The honest answer: the system worked at the volumes it was designed for; the market did not arrive at those volumes until after deregulation; and the system broke before the market caught up. But "worked, then was outgrown" requires acknowledging "did not, at the volumes it was designed for, attract its intended traffic." The contrarian's deeper point — that the 1996 expansion and the AeroTrain represent \$2.9 billion in evidence that the design had to be progressively overridden — is the aggregate the rest of this paper has to argue against.

The Aviation and Transportation Security Act of November 2001 added the second incompatibility. The post-9/11 sterile area, defined under 49 CFR § 1540.5, is a regulatory construct that did not exist when Saarinen drew his first sketch. TSA's Checkpoint Design Guide requires walls adjacent to non-sterile areas to be a minimum of 8 feet high, with continuous perimeter enclosure from document checker through screening lanes. Saarinen's terminal was, definitionally, undifferentiated public space. Departing, arriving, and connecting passengers shared a single floor plate. The post-9/11 regime broke that geometry. MWAA's engineering response — a 121,700-square-foot security mezzanine inserted between the main hall and the AeroTrain platform in 2009, excavated into the basement at a documented cost of \$1.3 billion to preserve the catenary roof above it — was the operational workaround. It was also the moment the curb-to-aircraft sequence Saarinen had organized the entire building around stopped being a coherent spatial experience.

Why the compact terminal cannot be brought back

The compact single-building principle is the one this paper releases by name. The intuitive case is volumetric: IAD handled 29 million passengers in 2025 against a master plan target of 45 million by 2045, and no physically achievable single terminal can serve 154 gates. That is true. It is not the load-bearing reason.

The load-bearing reason is the structural inheritance. The Saarinen terminal is a cable-suspension building. The 1-inch suspension cables, spaced 10 feet apart, are pre-stressed by the dead load of the 1,792 panels they carry. The 40-foot column grid at the perimeter is the lateral system; there are no shear walls and no braced frames. The 150-foot column-free interior over a now-1,240-foot footprint is the gesture, but it is also the cage: the building cannot accept intermediate columns, cannot accept a modified roof line, and cannot accept reconfigured perimeter geometry without breaking pre-stress equilibrium. The 1996 SOM extension proved the structural system can be extended longitudinally, replicating the column spacing and cable geometry to add 320 feet at each end. It did not prove anything about the interior. Every interior program — check-in islands, security dividers, retail, signage, mechanical distribution — must work within or hang from a structure that cannot be modified to accommodate it.

A 2026 reinterpretation proposing to restore the compact-terminal experience at hub volume would need to either accept the inherited envelope and compress 45 million annual passengers into 1,240 feet (geometrically impossible), or build a new compact terminal somewhere else on the property (a different building, not a reinterpretation). The chief engineer's exact formulation: "A genuinely new Saarinen-philosophy building — one that reinterprets the principles of modal separation, compact terminal, mid-field flexibility, and civic arrival experience — is architecturally and regulatorily a new building, not a modification of the 1962 one." The structural type cannot be reorganized to serve the operation. In chief-engineer terms, that is a new-building question stated in the language of renovation. The principle must be released because the structure forbids the alternative.

What survives is what the structural envelope and the modern operation can both accept: the legibility of the passenger sequence, the flexibility of the airside, and the separation of passenger from ramp. The compact building goes. The principle behind it — that a passenger should be able to understand where they are at every moment of the journey — is the principle worth carrying forward, expressed in 2026 idiom.

Where the philosophy still holds

What survives is more useful than what does not.

The first surviving principle is modal clarity, and from the operator's chair this is a question with a measurable answer. As of Concourse E's opening this September, MWAA will be running five distinct passenger-to-aircraft connection types simultaneously: jet bridge from the main terminal Z gates; AeroTrain to Concourses A and B; AeroTrain to Concourse C via the 500-foot underground moving walkway from the C-gate platform to the concourse it is supposed to serve; AeroTrain to new Concourse E; mobile lounge to Concourse D and to the International Arrivals Building. Five systems, five passenger experiences, each accumulated as the airport responded to immediate pressure, none designed against any organizing logic. Saarinen would have recognized this as the exact condition his terminal was built to prevent. Whether he would have endorsed the AeroTrain as the alternative is a different question, addressed below.

The second surviving principle is midfield flexibility — but read carefully. Saarinen's flexibility lived in the apron: aircraft could park anywhere because the lounge would find them. Once jetbridges replaced lounges, that specific flexibility evaporated. But the underlying principle — that the building should accommodate the operational form the airline industry will demand in 2055, not just the form it demands in 2026 — has never stopped being correct. The 2010 AeroTrain proved that subgrade engineering can be done under a live Saarinen terminal at a known cost (\$1.4 to \$1.5 billion for 3.8 miles of guideway plus the main terminal station, which required severing the building's foundation plane and rebuilding it from below while the building remained operational above). That cost is the floor for any further below-grade work, not the ceiling.

The third surviving principle is the civic-gateway argument, and it has to be defined once with operational specificity rather than invoked. By "civic gateway" this paper means a building that does four things measurable in observation. First, it presents a spatial threshold legible on entry — the passenger walking in from the curb knows they have entered the building, not just stepped under a roof. Second, the building's volume and light are organized so that the arrival sequence — from the curb to the first decision point — reads as a single intentional moment rather than as a corridor delivery to a checkpoint queue. Third, the building serves a representational role beyond its operational one: it appears in the city's self-image, is photographed, is identified as the place rather than as a terminal. Fourth, the international-arrivals sequence reads as a transition between systems of governance — the visitor lands and the building expresses that they have arrived somewhere specific. Foster's Hong Kong International, opened 1998 and now extended through Terminal 2 to a designed 100 million annual passengers, is the operational comparator that grounds each of these four. So is Kansas City's 2023 single terminal at a different scale: a building where the volume on entry, the sightlines to gate, and the absence of mode-switching deliver the civic threshold without any catenary roof or mid-century quotation. The Saarinen terminal in 1962 did all four. The Saarinen terminal in 2026, after the security mezzanine excavation and the Silver Line surfacing, does perhaps the third.

Saarinen wrote in 1962: "Federal architecture is static, but a jet-age airport should be essentially non-static, expressing the movement and the excitement of travel." The terminal was conceived as the international gateway to the American capital, not as a commercial concession platform.

Foster's Hong Kong is the cleanest contemporary parallel. The design integrates check-in, security, retail, and gate access under a single roof with a legible spatial hierarchy: clear lines of sight, minimal level changes, a vaulted roof that provides directionality, and a passenger sequence that reads as a single thing from curb to gate. The building is the concourse. The Saarinen-philosophy model — integrated experience, minimal modal breaks, the building as the spatial argument — is achievable to very high volumes within a single building, but only if the building is designed as a concourse from the start, rather than designed as a terminal and then connected to concourses. The 1996 SOM extension at IAD did the opposite: it extended the terminal longitudinally without engaging the question of what the extended terminal was for, and the midfield concourses were built

as a separate world. Foster designed the concourse and called it a terminal. IAD designed a terminal and added concourses. That difference is the difference between a building that works at hub scale and a building held together by a \$1.5 billion underground people-mover thirty-six years later.

The AeroTrain as the structurally cleaner expression of the surviving principles

A claim that needs its qualifier first. The observation below is about structural alignment of the AeroTrain with three of Saarinen's documented principles. It is not a claim about what Saarinen would have endorsed. Saarinen explicitly refused portable design philosophies; he insisted every problem required a unique solution. Any sentence beginning "Saarinen would have loved" is one this paper will not write. The deeper point cuts the other way: the most Saarinenian response to IAD in 2026 is not to revive 1962 principles but to spend a year with a stopwatch at the current airport — TSA queues, CBP, United's bank choreography, the AeroTrain platforms, the IAB, the curb at peak — and design something that responds to the problem as it exists now. Section V engages that honestly.

The observation here is narrower: on the three operational principles the documentary record supports, the AeroTrain expresses them more cleanly than the mobile lounge ever did. Modal clarity: the AeroTrain offers a single underground spine with timed headways; the lounge offered a single surface mode that became one of five over time. Midfield flexibility: the AeroTrain serves any concourse the guideway reaches and the master plan extends it to all of them by 2033; the lounge required apron paths and remote-stand availability that hub-bank operations stopped permitting. Grade separation: both modes separated passenger from ramp, but the AeroTrain does so without a vehicle on the active airfield.

From the carrier's chair, the airline-commercial-strategist reaches the opposite verdict on a different axis: "The plan effectively concedes that Saarinen's surface-level, mobile-lounge-mediated modal architecture cannot scale to the connecting-time requirements of a 2026 hub. The underground spine is not a refinement of Saarinen's concept — it is a replacement of its most distinctive operational feature." That is correct as a statement about the vehicle. It is also correct that the principles the vehicle served are not the same as the vehicle. The strategist and this paper are arguing past each other on the noun: replacement of the vehicle, extension of the principles. Both can be true.

The COO frame captures the synthesis: the AeroTrain is the correct architecture, improperly completed. The plan is right about modal clarity, wrong about treating the existing partial implementation as adequate; the spine is the discipline, but only if extended to every concourse and designed as the passenger sequence rather than as transit infrastructure with a finish package.

The Silver Line moment

There is a specific moment, datable to November 2022, when IAD stopped being a building you arrive at and became a building you surface inside of. The Silver Line

station opened that month, 1,150 feet from the front door of the main terminal, above grade, next to Parking Garage 1, connected to the terminal by an existing pedestrian tunnel that surfaces in the baggage claim level. The underground option that would have brought the station directly into the Saarinen sequence was \$640 million more expensive. The Virginia State Historic Preservation Office raised concerns about the visual impact of the above-grade alignment. MWAA chose the cheaper option.

The Saarinen-designed arrival sequence was: drive from downtown Washington, exit at the elevated curb, enter the main terminal at grade, experience the full volume of the building, board the mobile lounge. The 2022 transit sequence is: board Metro downtown, travel thirty-five miles, surface above grade next to a parking garage, walk through a baggage-claim tunnel, enter the terminal from below. The experience of the building is never encountered from the front. A passenger arrives via the utility side.

A reader sharper than v1 will ask the modal-split question. The current Metro share of IAD ground access [UNVERIFIED — HUMAN REVIEW: Stage 1 briefs did not source a specific percentage; field-verified rideshare/transit modal data should be sourced from MWAA ground-access statistics before publication] is a minority of total passenger arrivals; the overwhelming majority of passengers still arrive by car. The Silver Line decision is not the most consequential change in the arrival sequence in a generation if measured by passenger volume. It is the precedent for the next forty years of arrival decisions — the most recent capital decision in which the cheaper option won against the spatial logic of the building, and the conversation about what was being traded happened in a Section 106 review about visual impact rather than as a program-of-requirements decision about what the building is doing for the city. The \$22 billion plan is poised to make the same kind of decision at fifteen times the cost on every major element. The Silver Line is the practice run.

The \$22 billion plan, read on two axes

What the May 2026 MWAA plan does, stripped of the politics that surround its disclosure: it eliminates mobile lounges by extending the AeroTrain to every concourse; it builds four new linear concourses with jet-bridged gates; it demolishes the temporary Concourse C/D; it preserves the Saarinen main terminal and extends it 300 feet east and 300 feet west, replicating the SOM-replicated catenary; it commits \$21.8 billion in new bonds and \$1.1 billion in airport fees against a 2034 target completion.

The frame this paper uses is a two-axis decomposition. The first axis is operational coherence: does the plan resolve the modal-fragmentation, throughput, and connecting-time problems the operation actually faces in 2026? Throughput numbers, mode counts, connection times, and minimum gate-count requirements live here. The second axis is interior experience: at the moment a passenger is inside the building, does the spatial sequence read as a designed thing, or as a sequence of operational compartments connected by corridors? Sightlines, ceiling heights, dwell quality, daylight, threshold legibility, and wayfinding-without-signage-redundancy live here. The axes are independent because an operationally coherent plan can produce an interior that reads as a corridor (Hartsfield's underground transit mall, functionally), and an interior with

strong spatial qualities can be operationally incoherent (the 2022 IAD with five modes and excellent main-hall volume).

Read on these axes, the plan is faithful on operational coherence and at risk on interior experience. The AeroTrain extension finally resolves the five-mode fragmentation; the elimination of the surface-vehicle dependency closes the safety gap exposed in November 2025; the linear concourses with contact gates serve hub-bank operations that Saarinen's 1962 brief did not contemplate but that are not optional in 2026. On the second axis, the plan does not yet engage how the 300-foot terminal extensions are organized internally, what dwell quality the AeroTrain station platforms deliver, how the sterile-area boundary is spatially expressed, how the preserved main terminal connects to the new concourses, or whether the building still does any civic work — measured against the four-part definition above — beyond serving as a screening funnel and a baggage claim hall. The first axis is approximately right. The second is not yet addressed at all. The 1996 SOM expansion is the precedent for what happens when the second axis is left undefended.

The 1996 SOM precedent, used at structural strength

In 1996, SOM extended the Saarinen terminal from 600 to 1,240 feet, replicating the catenary structure, the column spacing, the concrete finishes, the terrazzo floors, and the precast panel system. By every formal and material measure, the work was faithful. During the same period, MWAA built Concourses A, B, C, and D in the midfield, with fixed jetbridges, abandoning the modal-clarity diagram entirely. The geometry was preserved. The operating logic was abandoned. The building looked like Saarinen. The airport began to operate like Atlanta. The operations-analyst's structural observation is the sharpest framing: "The Saarinen-design form was doubled in size; the Saarinen-design intent was inverted."

The risk in the \$22 billion plan is the same pattern at fifteen times the cost. Preserve the Saarinen shell, extend it 300 feet on each side in homage, run an underground spine to four new linear concourses, and accept an operational airport of 2034 that aspires to function like Hartsfield-Jackson. The Hartsfield comparison is an observer's reading of where the plan trajectory lands, not a quoted MWAA aspiration. A 2026 master plan for the international gateway of the American capital that lands operationally at Hartsfield without anyone having argued about what the building is doing civically is the SOM 1996 pattern at scale. The form will be preserved. The intent will be erased. The reader who has any feeling for what Saarinen was arguing about at Chantilly in 1959 should stop and consider what it means that this trajectory is the default unless something interrupts it.

The PIT comparison, with a verdict

The structural risk any honest reading of the \$22 billion plan owes the reader is the Pittsburgh comparison. PIT opened its \$1 billion hub terminal in 1992, designed at US Airways' specifications for a connecting hub. US Airways carried approximately 80% of PIT's traffic. US Airways began dehubbing in 2004; PIT enplanements fell from 9.15 million in 2002 to 3.97 million in 2013, a 57% decline. The airport boarded up 12 of 25 gates in Concourse B; Concourse E, which had 22 gates, became surface parking. PIT ran

a \$40 million annual deficit within two years of the hub collapse. The same pattern played out at CVG. Delta peaked there in 2005 at more than 600 daily flights, nearly 150 destinations, and 22.7 million annual passengers. By 2013, CVG had fewer than 180 daily departures, fewer than 6 million passengers, and a credit rating drop from A1 to Baa1.

The exposure profile this paper carries forward as the IAD analog is the strategist's compilation from the airport-CEO and airline-commercial briefs: United at roughly 70 percent of landed weight (the figure that drives MII voting under the Use and Lease Agreement) and at approximately 58 percent of enplanements, with the aviation historian's underlying framing the sharpest summary of what this means structurally: "The airport is not an MWWA facility that United uses. It is, functionally, a United facility that MWWA operates." The \$22 billion plan designs four linear concourses specifically around United's connecting-bank requirements. By the structural logic that produced PIT and CVG, IAD's exposure is comparable.

The honest verdict this paper commits to, where v1 punted: on balance, IAD is moderately less exposed than PIT was, but the difference is one of degree rather than kind, and the \$22 billion plan moves IAD in the wrong direction on this risk. The differentiators are real. United is structurally different from late-stage US Airways: the largest US-flag international carrier by available seat miles, not a financially distressed regional consolidator. The Washington traffic base is larger and more diversified than Pittsburgh's was. The transatlantic joint venture with Lufthansa and Air Canada creates a gateway dependency that CVG never had. The DCA slot constraint directs medium- and long-haul flying to IAD by regulation, not by airline preference. The diplomatic-arrival and federal-civic functions have no equivalent at PIT or CVG. These differences mean IAD's probability of a PIT-style outcome is materially lower.

But "lower" is not "low," and the plan does not currently hedge. The four-linear-concourse architecture, designed to United's banking choreography, makes the building harder to repurpose if United's network strategy shifts. The asset life of the program (forty years) is longer than United's longest planning horizon by a factor of four. Any of three scenarios — a future merger that consolidates United's network through a different hub, a transatlantic JV restructuring that reduces IAD's gateway role, a federal-policy shift that loosens DCA slots — would expose IAD to the structural pattern that produced PIT's surface-parking concourse. The plan needs to be designed so that a meaningful non-United operator could absorb 20 to 30 percent of the gate capacity without structural modification. That is what hedging looks like at the program-of-requirements level. Move 7 specifies it.

IV. The Counter-Case, Honestly Presented

The strongest case against everything just argued begins with operations, not aesthetics.

The contrarian's lead number is the one this paper now puts at the head of the counter-case. In 1975, Dulles served 2.5 million passengers against a 6 million design capacity —

42 percent utilization — while DCA served 11.7 million. The mobile-lounge airport, with its design philosophy fully intact and its operational concept unviolated, ran at less than half capacity in the era it was supposed to dominate, in a market where its principal alternative ran at more than four times its traffic. By 1985 the original design, in the airport authority's own planning record, "was no longer well-suited to Dulles's role as a hub airport." By 1987 it had been operationally relegated. The 1996 expansion and the AeroTrain represent \$2.9 billion in evidence that the design had to be progressively overridden. This is not a design philosophy that aged well. It is a design concept in a sixty-three-year holding pattern that the institution has been spending against to keep operationally viable.

The case continues with the refusal of the framing itself. The thesis claims to separate "design philosophy" from "aesthetic." But Saarinen's design philosophy *was* his aesthetic. The catenary roof at Dulles is not decoration. It is structural, thermal, and psychological simultaneously. The argument is the curve. To say one can take the philosophy without the look is to say one can take Miles Davis's musical thinking without the silences. The silences are the music. "Design philosophy separable from aesthetic" is the definitional move of sophisticated nostalgia — it launders a preference for the 1962 building through the language of principle.

The deregulation gap sharpens the same point. Saarinen designed for a CAB-regulated, point-to-point world. The Airline Deregulation Act of 1978 restructured the industry around hub-and-spoke networks sixteen years after the airport opened. Applying a 1962 design philosophy to a 2026 connecting hub is a category error. Foster's Hong Kong delivered monumental civic scale, wayfinding clarity, modular expansion, and a legible passenger sequence without mobile lounges, without mid-century romanticism, and without requiring a design-philosophy argument rooted in a specific architect. Kansas City's 2023 single terminal resolved the modal-clarity problem on time and under budget, with no reference to Saarinen. The principles the thesis attributes to Saarinen exist independently of him.

The case continues with Saarinen's own self-description, which is the thesis's most uncomfortable enemy. The 2005 *Journal of Architecture* paper "Heroicisation, compensation and coronation or the risks of reevaluating Eero Saarinen" makes the academic version of the argument. The paper's three nouns are the risks of a Saarinen revival in any form: heroicisation (treating an architect's biographical narrative as the carrier of architectural truth), compensation (using the revival to rectify a perceived historical underestimation, which substitutes corrective sentiment for analytic judgment), and coronation (institutionalizing the architect's status in a way that closes future critical readings). The paper's specific application targets exactly the move this paper makes: extracting a "design philosophy" from a specific architect, then deploying it as a normative framework for present capital decisions. Saarinen himself wrote what amounts to the same critique in advance: "Sometimes the problem and the times are right for an entirely new functional approach to a problem." If you take him at his word, the Saarinenian response to IAD in 2026 is not to reinterpret his 1962 design principles. It is to do at the current airport what his office did at the comparable airports of the late

1950s: a year of direct observation paired with quantitative comparison of program alternatives, paired with the willingness to let the building's form follow what the observation produces. The thesis's argument for Saarinen-philosophy reinterpretation may be the most anti-Saarinenian thing one could do.

The regulatory floor is not a design parameter to be worked around. It is a security-critical statutory framework enacted after actual failures. The post-9/11 sterile-area paradigm under 49 CFR Part 1542, ADA accessible-route requirements under 49 CFR Parts 37 and 38, and CBP's Airport Technical Design Standards collectively prescribe corridor geometries Saarinen's program brief never addressed. At every named deployment reviewed, biometric systems are added on top of existing checkpoint architecture, not in place of it. The 8-foot wall stays. The hope that 2026 technology dissolves the architectural constraint is the hope that wishes the wall away.

The case ends with political economy. MWAA has been in oversight recovery mode for over a decade following the 2012 DOT OIG audit, which found the authority had used categorical exceptions to limit competition on almost two-thirds of contracts exceeding \$200,000. Permanent statutory DOT IG oversight was established in 2014 legislation. MWAA's coverage ratio is compressing toward 1.3x against a 1.25x covenant as the airport executes \$5.5 billion in new bond issuances; the projected CPE of \$90.64 by 2035 would put IAD well above the level at which United left as a hub the first time. The airport-CEO's framing: "the CEO who walks into the next ratings briefing with a Saarinen-philosophy argument as a binding capital constraint is the CEO who generates questions they cannot answer in a ratings call."

That is the strongest version of the counter-case. It is not a strawman. A reader who has not internalized it should not be persuaded by anything that follows.

V. Why the Counter-Case Is Insufficient

The concessions, by name.

Saarinen's design philosophy is not freely separable from his aesthetic. The catenary roof is structural. The compact terminal is a consequence of the lounge. The argument is the curve.

The mobile lounge as a specific vehicle is operationally obsolete and was warned to be obsolete from opening day. Any 2026 reinterpretation that proposed to revive it would fail the 35-minute minimum connection time, the ADA boarding tolerances, the FOD requirements on active taxiways, the safety record of the existing fleet, and the basic patience of any operator who has had to coordinate a hub bank around vehicle availability.

The single compact building as the organizing element of the passenger journey is obsolete for both reasons argued above: the operation cannot fit, and the structural type cannot be reorganized.

The Saarinen frame is not the only entry point to the principles the thesis cares about. Foster, KCI, Kansai, Madrid Barajas all do related work without invoking Saarinen.

The 1975 utilization gap is a real indictment of how the design performed in its own era. The 42 percent utilization against DCA's 11.7 million passengers means the philosophy did not, in the period when passengers had every chance to choose it, produce the market preference its defenders claim it was designed to produce.

The \$2.9 billion aggregate of subsequent override capital deserves its own response rather than its own concession. The contrarian reads it as capital spent overriding the operational concept while preserving the architectural envelope. The reading this paper offers in reply: the override capital divides into two categories. The 1996 SOM extension and the midfield concourses (the per-component costs are not separately published in the available sources, but the chief-engineer brief notes the SOM extension as a documented expansion premium and the midfield concourses were a separate capital wave; together they account for the larger of the two halves) were override capital that preserved form and abandoned intent — the building doubled in length, the modal-clarity diagram was discarded, the airport began to operate like Atlanta. The 2010 AeroTrain at \$1.4-1.5 billion and the 2009 security mezzanine at a documented \$1.3 billion were override capital that worked within the structural inheritance to restore a version of the modal-clarity principle the lounges had stopped delivering — an underground spine that re-centralized the passenger sequence and pulled the post-9/11 security mass below the catenary so the main hall would remain a main hall. One half spent against intent, the other half spent in service of it. The aggregate is also a forensic record that distinguishes the two kinds of override, and that distinction is the discipline the next \$22 billion needs. If the upcoming program is read as \$22 billion in additional override capital — and it will be, in twenty years, by someone writing this kind of paper — the question will be which half it joined. The claim is that the override capital that preserved intent is distinguishable from the override capital that preserved only form, and that the program of requirements is the moment that distinction can still be designed in.

The most damaging concession is the Saarinen-against-himself point. He refused portable philosophies. The most Saarinenian response to IAD 2026 is direct observation at the current airport, not the reinterpretation of 1962 principles. The *Journal of Architecture* paper's three nouns — heroicisation, compensation, coronation — apply to this paper's framing and have to be answered.

These concessions are load-bearing. They are why the counter-case is insufficient rather than wrong.

What the counter-case does not address: the program of requirements being defined, right now, for a \$22 billion capital decision that will determine the international gateway of the American capital for the next forty years. That definition is happening under political pressure, under bond-market constraints, under the residual leverage of a single anchor carrier, under the throughput optimization preferences of TSA and CBP, and

under the procurement timelines of a federal RFI with no confirmed funding mechanism. None of those forces is illegitimate. What is missing — what was present in 1959 in a way it is not present in 2026 — is anyone in the capital-decision conversation whose role is to ask, at full observational seriousness, what the building is for at the level of human experience, and to defend that answer through the value-engineering cycles, the airline negotiations, the TSA change orders, and the FAA certification process.

The Saarinen frame is useful in 2026 not because Saarinen had the right answer for 2026. He did not, and would not have claimed to. The frame is useful because it names the absence of the role he played in 1959. The architect with the measurement instrument. The person who treated the experience of arriving at an airport as a program requirement rather than a design preference. The person who held that line when the airline wanted more gates, the FAA wanted faster checkpoint flow, the contractor wanted simpler geometry, and Congress wanted lower cost.

The slacker's question deserves a direct answer rather than a deflection. The most Saarinenian thing MWAA could do in 2026 is spend a year with a stopwatch at the current airport. Section VI does not propose that. It proposes a 60-to-90-day passenger-experience study — a compressed version of Saarinen's empirical work, scoped to inform program-of-requirements decisions rather than to drive a new design, justified by a bond-issuance schedule the institution cannot move. The compression is a concession to the political-economic environment, not a methodological equivalent. A 60-to-90-day study is the methodological pre-condition for the moves that follow it; it is not a substitute for the multi-year stopwatch work the documentary record attributes to the Saarinen office. The thesis is that the compressed version still does work the present capital conversation does not currently do — but the honest framing is that it does less work than the original, by a factor that the bond schedule, not the strategist, determined.

The named failure mode of civic-monumental ambition with the observational discipline removed is the Santiago Calatrava World Trade Center Transportation Hub — the Oculus. Construction cost approached \$4 billion against an initial \$2 billion estimate; Calatrava was paid \$80 million; the operable skylight that opens once a year on September 11 cost \$32 million; the building is, in the published criticism, "infamously nightmarish when it comes to maintenance." It is primarily experienced as a shopping mall with a spectacular ceiling. The architectural ambition is genuine. The operational program was not the organizing question; the civic-monumental gesture was. The Oculus is what happens to a building when an architect with civic ambition meets a capital program with no countervailing observational discipline and a revenue model that requires retail to fill the sterile-side floor area. Every large terminal program in the last twenty years has been partly financed by concession revenue projections that require heavy retail in the sterile zone. Saarinen's single-compact-terminal concept produces exactly the kind of compressed, high-traffic sterile zone that concession operators want. That is not a Saarinen principle. That is a coincidence the \$22 billion planners should not mistake for intent. The Oculus precedent for IAD is specific: a \$22 billion program with no confirmed funding mechanism, with bond pricing that will require revenue commitments to underwrite, in a political environment that interprets monumental civic

gestures generously, is the exact conditions under which an Oculus-shaped outcome becomes the path of least resistance. The Saarinen-philosophy frame, in this application, is the discipline that names the failure mode in advance and asks what specific design moves prevent it.

So the conditions under which the thesis holds.

The thesis holds if MWAA reads Saarinen as a method, not as a form. The catenary stays where it is, the SOM extensions stay where they are, the 300-foot extensions go where the plan puts them. What changes is that the interior organization of those extensions, the dwell experience in the AeroTrain stations, the spatial expression of the sterile-area boundary, and the relationship between the preserved main terminal and the four new concourses are treated as program requirements with measurable success criteria — passenger sequence legibility, dwell quality, intuitive wayfinding, civic threshold experience at arrival — and not as design preferences to be traded for gate flexibility or construction schedule.

The thesis holds if the Section 106 Programmatic Agreement is used to embed those criteria. The VDHR/VASHPO has determined that a project-specific PA is appropriate. Whether it memorializes design-quality obligations depends on whether MWAA negotiates for them. The window closes when the board approves the bond issuance.

The thesis holds if MWAA can frame the philosophical question in operational and financial language that survives the bond-rating briefing and the United MII conversation. "We are reconsidering our capital philosophy in light of our founder architect's design intent" loses both conversations. "We are defining the program of requirements with measurable spatial and experiential criteria that will reduce passenger-experience risk on a \$22 billion asset, and we are documenting those criteria in instruments that survive value engineering" wins both. The Saarinen frame is shorthand inside MWAA. It is not the framing used outside.

The thesis holds if the Hartsfield default is taken seriously as an unintended consequence to be designed against, not as an aspiration. Hartsfield is a great hub. It is not what Dulles should aspire to be. The international gateway role, the diplomatic-arrival role, the federal-civic role at the capital are program elements that have no equivalent at ATL.

If those conditions hold, the thesis is correct. If they do not, the counter-case is correct, and the right move for MWAA is to execute the \$22 billion plan as a competent infrastructure program and accept that the international gateway of the American capital will operate like Hartsfield in 2034.

VI. Implications for MWAA

What follows is the deliverable. Each move names the live program element it addresses, the intervention, and the legal or contractual instrument through which the intervention

is made binding. The Programmatic Agreement is used across multiple moves because it is the live instrument with cross-cutting reach into the listed terminal and its directly connected envelope. Move 1 is a methodological precondition for Moves 2, 3, and 5.

Move 1: Commission a 60-to-90-day passenger-experience study before the program of requirements for the extensions and the AeroTrain platforms is locked

This is the operational instrument that makes the "stopwatch as philosophy" claim concrete rather than rhetorical. By "passenger-experience study" this paper means a defined, bounded scope: a 60-to-90-day observational study of named flows at IAD under existing conditions, conducted by an independent firm reporting to the MWAA board rather than to the program-management contractor. The named flows are specific. First, the curb-to-checkpoint-to-AeroTrain-to-gate sequence for departing originating passengers, timed and observed across morning and evening peak periods at the main terminal Z gates and across the AeroTrain transfer to Concourses A, B, and C. Second, the international-arrivals sequence at the IAB and at the Concourse C IAB extension, from aircraft door through CBP through baggage to landside, timed and observed across the widebody arrival peak windows. Third, the connecting-bank micro-sequences United operates between Concourses C, D, and (after September) E during the morning push and the late-afternoon push — gate-to-gate timing, AeroTrain platform dwell, walking distances, mode transitions. Fourth, the curb peak at departures during the Friday evening and Monday morning windows. Fifth, the Silver Line arrival sequence from station surface to terminal entry.

The output is a documented baseline of mode-transition counts, dwell-time distributions, walking distances, decision points where passengers visibly hesitate or backtrack, and wayfinding-without-signage success rates. That baseline is the empirical input that makes the spatial criteria in Moves 2, 3, and 5 specifiable rather than rhetorical.

Cost: low six figures against a \$22 billion program. Schedule: must be commissioned within 60 days of this paper's reception to complete before the program of requirements is locked at bond issuance. Instrument: MWAA board commission, reporting directly to the board's program oversight committee, not to the program-management contractor. The reporting line matters; the institutional history of this airport is that studies reporting to the PMC become inputs to the PMC's value-engineering preferences rather than independent reads.

The study is not equivalent to Saarinen's multi-year stopwatch work. Section V is honest about that gap. It is the compressed version that the bond schedule permits, and the methodological precondition without which the rest of this section is a list of preferences.

Move 2: Define the program of requirements for the 300-foot terminal extensions, and embed the criteria in the Section 106 Programmatic Agreement

The live program element is the 300-foot east and west extensions of the Saarinen main terminal in Phase 1 of the \$22 billion program (\$6.2 billion, construction start April 2027, target completion within the 2034 window). The structural envelope is fixed by the inherited catenary geometry; SOM's 1996 work proved the extension can be done with formal fidelity. What has not been defined is the interior organization. Is each extension a check-in hall? A sterile-side concourse extension feeding the AeroTrain spine? A flexible mixed-use volume? Absent a positive intervention, each extension will become whatever the value-engineering cycles produce when bond pricing closes.

The intervention is to define four measurable spatial criteria — sightline legibility from each curb entrance to the AeroTrain platform; single-decision wayfinding without signage redundancy for at least 80 percent of departing passengers based on the Move 1 baseline; dwell quality benchmarks at every mode transition specified against the same baseline; no more than one mode change between curb entry and security — and embed them in the project-specific Programmatic Agreement now being negotiated with the Virginia SHPO. The PA is a legally binding document under the National Historic Preservation Act, not an advisory opinion. Criteria embedded in it survive the value engineering. Criteria not embedded in it do not.

CPE impact: approximately neutral. The criteria do not require additional volume; they require disciplined organization of volume already in the program. Offsetting savings come from reduced post-occupancy retrofit, which the airport's institutional history suggests is the dominant component of long-run capital cost on this site. Instrument: project-specific PA, in active negotiation, executed before the FAA's Finding of No Significant Impact or Record of Decision on the EA.

Move 3: Make the AeroTrain station dwell spaces — including the planned extension stations — design-quality elements in the PA

The live program element is the \$3.75 billion AeroTrain extension to new Concourses D, B, and the central spine connector, with construction targeted to begin January 2028 and complete December 2033. The 2010 SOM AeroTrain station under the main terminal already does design-intent work at the architectural level: faceted concrete-beam roof, skylights, terrazzo flooring referencing the original terminal. The intent is correct. The execution stops at the architecture and does not extend into the platform dwell experience itself — a passenger waiting two minutes for an AeroTrain in 2026 is in a transit station that could be anywhere. The extension is the opportunity to define dwell quality, spatial generosity, natural light penetration where geotechnically possible, and architectural continuity with the Saarinen idiom at the new stations.

The intervention is to specify three program-of-requirements line items in the AeroTrain extension design package: a minimum platform ceiling height set at a design-quality premium above the TRB-typical 12-foot APM platform, with the exact dimensions

calibrated by the Move 1 dwell-quality findings; daylight provision at each station via either skylight penetrations or fiber-optic light tubes where direct apertures are not feasible; and surface-material continuity with the SOM 2010 station's terrazzo-and-cast-concrete palette specified in the construction drawings rather than treated as a finish-package decision.

Against a \$3.75 billion budget, the design-quality finish premium is rounding error at the program scale and consequential at the per-station scale — the line items above are not what determines whether the program is affordable; they are what determines whether the stations read as designed spaces or as transit interludes. Instrument: PA design-review provisions for all stations within the listed terminal envelope and for the design relationship between the listed envelope and the directly-connected new stations, applied uniformly across the AeroTrain extension.

Move 4: Lock the demolition date for Concourse C/D in the Use and Lease Agreement amendment that accompanies the bond issuance

The live program element is the demolition of Concourse C/D, currently scheduled to occur once the new Concourse B opens in the 2028 to 2030 window. C/D was built in 1983 as a temporary structure. It has been temporary for 43 years. It has received approximately \$100 million in passenger-boarding-bridge and systems capital improvements (the documented program covered HVAC, plumbing, and electrical relocation for the PBBs, on top of separate sanitary and HVAC rehabilitation cycles). It is United's primary hub concourse and the daily operational liability the airport COO describes — low ceilings, replaced HVAC, sanitary system reconfigured after flooding, jet bridges retrofitted for aircraft larger than the original gate design. The institutional pattern at IAD is that demolition dates slip because the demolition is never locked before the replacement opens.

The intervention is to lock the C/D demolition date in the Use and Lease Agreement amendment that will accompany the \$22 billion bond issuance, with United as the affected party formally signed up to the schedule. The instrument is the U&L Agreement amendment, which under the January 2025 hybrid residual structure requires majority-in-interest approval that United alone controls. The negotiation should pair the demolition date with United's interest in the new Concourse B's banking choreography, which is the airline's reason to support the schedule. CPE impact: marginally positive over the long run (eliminates the \$2.5 million annual operating cost of the retained PTV fleet and the sustained-life capital cycle on C/D), neutral in the bond-pricing window. Without the lock, C/D becomes the next IAB — a "temporary" facility that operates for another two decades, with United's premium passengers continuing to experience low ceilings and submarine acoustics.

Move 5: Treat the new midfield FIS facility as the single most important design decision in the entire program, and define its experiential criteria in the PA

The live program element is the new Federal Inspection Services facility in the midfield concourses, displacing the 1991 International Arrivals Building. The IAB processed 3.81 million passengers in fiscal year 2024 and ran 348 days with peak wait times exceeding 60 minutes during the April 2022 to March 2023 reporting period, including 98 days exceeding two hours. Locating the FIS in the midfield concourses rather than in a separate building is, structurally, the most aligned move in the entire plan with what Saarinen's principles would produce in 2026 idiom: bring arrivals processing to where the aircraft are, rather than moving passengers across the apron to a separate facility.

The risk is that the FIS is treated as a CBP procurement requirement satisfied at minimum specification. The Customs ATDS prescribes the corridor sequence and the inspection geometry, but it does not prescribe dwell quality, light, ceiling height, the spatial expression of the transition from arrival corridor to baggage hall, or the way the connecting-passenger recheck loop reads against the terminating-passenger exit loop. These are the elements that determine whether a passenger arriving in the American capital from Frankfurt experiences a building that feels designed, or a CBP-spec corridor in a sheetrock box.

The architecturally-defended version is specifiable on five axes. First, the arrival corridor from aircraft door to primary inspection: ceiling height above CBP's minimum spec (which permits 9-foot acoustic-tile ceilings), daylight penetration via clerestory where the airside roof envelope permits, and a sightline from the aircraft door to the first decision point that delivers the international visitor's first read of the building. Second, the primary inspection hall: ceiling height that registers the volume change from the corridor — the architectural moment when the visitor knows they have arrived somewhere specific — and a queue geometry that does not force the visitor to face a wall. Third, the baggage hall: daylight, ceiling height matched to the inspection hall, and sightline legibility to the customs exit and to the connecting-passenger recheck route. Fourth, the connecting-passenger recheck loop: a separated route that does not return connecting passengers to the public side of the FIS, with a recheck queue dwell quality calibrated to the United MCT requirement and observed during the Move 1 widebody arrival peak. Fifth, the spatial expression of the transition from CBP-controlled space to public-side US territory: a designed threshold, not a glass door with signage. These criteria are additive to the CBP POR rather than substitutive — CBP's requirements are mandatory; MWAA's design-quality criteria specify how the same square footage is organized. The downstream operational savings from a functional FIS — reduced misconnect costs to United, reduced ground time for inbound widebodies, reduced staffing costs on connecting-passenger handoffs — are quantifiable and accrue to the airline rate base, which is the commercial argument for the PA-embedded criteria's survival through value engineering.

Instrument: PA design-quality obligations layered onto the CBP POR. CPE impact: built into the existing program scope. This is the move where the Saarinen-method discipline matters most — the FIS is the building's first impression on every international visitor to the American capital for the next forty years, and the CBP-spec default is a building that fails the civic-gateway test on all four criteria defined in Section III.

Move 6: Negotiate design-quality obligations into the Use and Lease Agreement amendment for the new concourses, on the airline-supported axis only

United's interests in design quality are asymmetric. United supports design moves that benefit branded passenger flow, premium-cabin product space, wayfinding clarity, gate-bank coherence, and connecting-itinerary legibility. United resists design moves that suppress branded terminal differentiation, that require remote-stand operations in the name of modal purity (incompatible with 35-minute domestic MCT), that degrade gate-bank adjacency, or that raise CPE without proportional network benefit.

The intervention is to bring design-quality criteria into the United MII negotiation only on the supported axis: gate-bank legibility, connecting passenger sequence quality, premium-product spatial integration, and wayfinding standards measurable as connection-completion rate against the Move 1 baseline. These criteria show up as misconnect-cost reductions, premium-cabin yield retention, and lounge-eligible customer satisfaction metrics — all of which improve United's hub contribution margin. The Saarinen-philosophy criteria unaligned with United's interests belong elsewhere. The U&L Agreement amendment accompanying the \$22 billion bond issuance is the instrument; the negotiation should pair United's support for these criteria with the C/D demolition lock from Move 4 and with the Concourse E United Club arrangement already in place. CPE impact: criteria-dependent, but the United-aligned axis is approximately CPE-neutral or favorable because the airline absorbs the design quality as commercial value.

Move 7: Commit to a hedging design against the residual PIT-style anchor-carrier risk

The structural risk any honest reading of the \$22 billion plan owes the board is the Pittsburgh comparison. The verdict argued in Section III: IAD's exposure is moderately lower than PIT's was, but lower is not low, and the four-linear-concourse design moves the airport in the wrong direction. The hedge is not a question of whether to build the four concourses; it is a question of whether they are designed to be operable by a meaningfully different carrier mix.

The intervention has three elements. First, design the new Concourse D (non-United domestic, \$3.7 billion, construction 2027 to 2033) for swing capacity — common-use gate infrastructure, fueling and baggage system architecture that does not assume a single carrier's bank choreography, gate sizing that accommodates both narrow-body and wide-body international operations. Marginal construction cost above preferential-

use design is real but bounded; it is consequential at the gate-program scale and rounding error against the \$22 billion envelope.

Second, the carrier-concentration cap: no more than 70 percent of new gate capacity created by the program held under preferential use by a single carrier at any point in the asset life. United at roughly 70 percent landed weight is the airport's current reality; the program should not bake in a path that makes the percentage higher and irreversible. The instrument question is real: this is the category of move United resists. Move 6 carved United-resisted criteria out of the U&L Agreement on principle. Move 7 cannot follow the same logic on this specific cap, for one reason: the PA's jurisdiction is the listed historic envelope, and the carrier-concentration cap applies primarily to the new midfield gates, which are outside the PA's reach. The cap must go into the U&L Agreement to be binding — there is no PA-shaped instrument available for it. That is the honest tradeoff. The cap is harder to land in the U&L Agreement than a United-aligned design criterion would be, and the negotiation should pair the cap with United's interest in the Concourse E lounge arrangement, the C/D demolition schedule, and the Concourse B gate-bank specification. If the cap cannot be landed, the program-management alternative is to memorialize the cap as a board-adopted master plan policy with reporting requirements rather than as a U&L covenant — non-binding but visible, which is the second-best instrument when the binding one is not achievable. The strategist's preference is the binding version; the strategist's honest read of the United MII conversation is that the non-binding version may be what survives.

Third, scope the Airport Terminal Program grant application narrative to emphasize the multi-carrier-resilience element of the design, which scores against the program's capacity-and-access criteria and creates a federal record that the design intent is not single-carrier-optimized.

MWAA staff who frame this as a multi-carrier-resilience design move rather than as a hedge against United have the better political path to United's MII approval — the airline can support a design that does not constrain its own growth while accommodating the airport's risk management.

These seven moves do not require MWAA to invoke Saarinen by name in any public document. They do not require fighting TSA on sterile-area architecture, a fight not worth having on the time horizon that matters. They do not require revisiting the 2025 to 2039 Use and Lease Agreement, which represents the most important financial achievement of the Potter era — the moves work through an amendment that accompanies the \$22 billion bond issuance, not through a reopened base agreement. They do not require slowing the \$22 billion program.

They require treating the program of requirements as a program of requirements rather than as a procurement specification, defining experiential criteria with the same rigor as throughput criteria, and documenting those criteria in instruments — the project-specific Programmatic Agreement, the Use and Lease Agreement amendment, the

Airport Terminal Program grant scope — that survive value engineering. They require the empirical baseline (Move 1) that makes the criteria specifiable rather than rhetorical. They require asking the deferred PIT question on the record, before the bond issuance commits the scope. They require, at the FIS facility specifically, treating the building's first impression on every international visitor to the American capital as a program element rather than a CBP-spec output.

VII.

The terminal has operated for sixty-four years. It has been doubled in length to the dimensions originally specified, has had its operating diagram replaced and partially restored, has buried a 121,700-square-foot security mezzanine in its basement at a documented \$1.3 billion and an automated people mover under that at \$1.5 billion, has watched its modal-clarity principle accumulate five parallel passenger-to-aircraft systems where it once contained one, and is now the live framing for a \$22 billion capital decision being made by people for whom the stopwatch work that produced the building is an artifact in a documentary.

Saarinen will not be in the room. He refused portable philosophies; he insisted every problem requires a unique solution; his Dulles design produced a 1975 utilization of 42 percent against DCA's 11.7 million passengers in the same year; his TWA terminal closed in 2001 after thirty-nine years of operational disqualification. The hagiographic case for invoking his name is weak and was weak in 2005 when the *Journal of Architecture* paper made the academic argument against it. This paper has not made the hagiographic case. The case it has made is narrower and less flattering to the building it is defending: the observational discipline that produced the 1962 terminal at the operational level has, in 2026, no equivalent inside MWAA's \$22 billion decision. The empirical baseline the documentary record attributes to the Saarinen office — a year of direct observation of passenger behavior at comparable airports, paired with quantitative comparison of more than 100 program alternatives — is not present in the capital conversation in any structured form. The named failure mode of its absence is the Calatrava Oculus, which is a published case study in what civic ambition without observational discipline produces. The published case is not yet IAD's. It can become IAD's, by default, if the program of requirements for the 300-foot extensions, the AeroTrain platforms, the FIS facility, and the relationship between the preserved main terminal and the new concourses is left to be defined by value engineering at bond pricing.

The instruments exist. The Programmatic Agreement, the U&L Agreement amendment, the Airport Terminal Program grant scope, and the 60-to-90-day passenger-experience study are the four levers that, used in the next twelve months, would bind the program of requirements to measurable experiential criteria before the bond issuance closes the design conversation. The verdict this paper commits to: by the date MWAA's board commits a bond issuance to a fixed scope — currently targeted within the next twelve to eighteen months — the institution will either have used those four instruments to embed observational discipline into the program of requirements, or it will have deferred the

question to the value-engineering cycles that have, in the documentary record, produced most of the major capital misses in the airport's sixty-four-year history. The first outcome leaves the next forty years of arrivals at the American capital's international gateway designed against measurable spatial criteria. The second leaves them designed against the path of least resistance at bond pricing. There is no third outcome. The window closes when the board votes.

What is on the table is not Saarinen's building. It is what the next building does for the city, measured against the four civic-gateway criteria, observed at the curb at 6 a.m., timed at the FIS at the widebody peak, walked from the Silver Line platform through the baggage tunnel to the main terminal floor. The stopwatch is still the instrument. The question is whether anyone in the room is holding one.

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